EL NIÑO ACTION BRIEFING · 30 OCTOBER 2026

Kafouri, Khartoum North

El Niño's heat here is mostly past. What it brings now is a failed harvest and higher food prices into 2027, while cholera and mosquitoes kill first. Make drinking water safe and put oral rehydration salts within reach of every square.

PROVINCE	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Khartoum State (province level)	1,100 forecast (970 of them in October); 6,200 already past (Jun–Sep)	6-month forecast, Oct 2026 – Mar 2027	Cholera, dengue, and malaria; food prices into 2027; heat again from April

16/16 recommendations checked. 12 corrected, 1 unverified (dropped), 6 contraindicated (the six "Do not do this" cards). Exposure figure: Climate Impact Lab El Niño Heat Mortality Tracker data, version of 26 October 2026, pre-peer-review, methods forthcoming. It is published for the whole of Khartoum State, not for Kafouri, and it counts heat only, so it cannot see cholera, mosquito-borne disease, or hunger, which are this place's main risks before the peak. The funding target fell from the panel's SDG 1.72 billion (US$195,000) to SDG 968 million (US$110,000) after verification (−44%): a cholera vaccine round cannot be ordered until an outbreak is confirmed, so it moved to the ceiling, and sorghum prices were re-estimated. Kafouri's population is not published, so per-person costs use a 50,000-person planning assumption.

EXECUTIVE SUMMARY — AT A GLANCE



15.64°N 32.57°E, Sudan · Natural Earth

- **The risk.** The heat deaths have mostly happened (6,200 of 7,300 by September); now the driest rains in decades push food prices up into 2027, on top of cholera, dengue, and malaria in a city with broken water and power.
- **For you.** From today, treat all drinking water with a chlorine tablet or 2 drops of bleach per litre, and keep oral rehydration salts at home.*
- **The money.** SDG 352 million (US$40,000) floor, SDG 968 million (US$110,000) target, SDG 1.87 billion (US$212,000) ceiling; chlorine and rehydration points through the Bahri Emergency Response Room buy the most lives.
- **Do not.** Do not clear rubble or move metal objects before a block is surveyed: 23 people, 21 of them children, were killed or injured by explosives in Khartoum State in January–March 2026.

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* Suggested formulation: check with a health worker before sharing.

FOR YOU, AS A RESIDENT 9 STEPS

For a household in Kafouri, this El Niño arrives less as heat than as dirty water, mosquitoes after the rains, and food that costs more every month into the 2027 lean season, in a district where the power is off 8–10 hours a day and the pound loses value each week. The ranked actions below are written for the State Ministry of Health, the Bahri Emergency Response Room, and their partners. These nine are the ones you can take yourself, grouped by what they protect. Prices are estimates: no Bahri retail prices are published.

1 · Water: from today, treat every drop you drink, and keep it covered *

PARTIALLY CONFIRMED FROM ACTION 2

BY WHEN	COST TO YOU	PROTECTS
Now, every day	Tablets are free from aid groups; bleach price not published	Everyone who drinks at home

Use the chlorine tablets that aid groups hand out first, because bleach strength in Sudan varies. Without tablets, use plain unscented household bleach of 5–6%: 2 drops per litre, or 10 drops per 20-litre jerrycan, doubled if the water is cloudy, then wait 30 minutes before drinking (US CDC). Stronger bleach needs fewer drops, so read the label. Store water in a covered jerrycan and pour it out rather than dipping a cup in. Treating water at home cut child diarrhoea by about a third across 25 studies in low- and middle-income countries (Wolf et al., *Lancet* 2022). The Bahri water plant has only partly resumed, and the 2025 cholera wave in Khartoum followed the power cuts that stopped the pumps. *

2 · Cholera: keep oral rehydration salts at home, know the home recipe, and take the vaccine when it is offered *

PARTIALLY CONFIRMED FROM ACTIONS 1 AND 3

BY WHEN	COST TO YOU	PROTECTS
15 Nov 2026	About SDG 530 (US$0.06) a sachet at UNICEF's price; pharmacies charge more	Children and older people with diarrhoea

Keep ten sachets of oral rehydration salts (ORS). Without them, mix 6 level teaspoons of sugar and half a level teaspoon of salt in 1 litre of clean water; keep the word "level", because too much sugar makes diarrhoea worse. Start it at the first watery stool. If someone cannot drink, or the diarrhoea is very heavy, go to a clinic at once: treated cholera rarely kills, while untreated severe cholera can kill

* Suggested formulation: check with a health worker before sharing.

within hours (WHO). In August 2026 doctors reported uncounted cholera cases in Al-Ezba, next to Kafouri. If a vaccination team comes, take the dose: one dose protected about 87% of people in the short term in Juba, South Sudan (Azman et al. 2016).*

3 · Mosquitoes: every Friday, empty, scrub, and cover every water container, and sleep under a net

PARTIALLY CONFIRMED FROM ACTION 4

BY WHEN	COST TO YOU	PROTECTS
Start by 15 Nov 2026, weekly to March	None for the sweep; a net about SDG 17,600 (US$2) before delivery	The household, from dengue and malaria

The dengue mosquito breeds in stored water: barrels, clay pots (zeer), tanks, tyres, and plant pots. In Port Sudan about half of drinking-water containers held larvae. Scrub the sides, because the eggs stick there, and put a lid or a cloth over every container. Medics counted 186 dengue deaths in Samarab, in Bahri, in October 2024. Nets protect against malaria, not dengue: the dengue mosquito bites by day. The invasive malaria mosquito *Anopheles stephensi*, which also breeds in water tanks, has been found in Khartoum.

4 · Food: in December or January, buy grain rather than holding pounds, alone or with your square

PARTIALLY CONFIRMED FROM ACTION 5

BY WHEN	COST TO YOU	PROTECTS
Dec 2026 – Jan 2027	About SDG 270,000–360,000 (US$31–41) for a 90 kg sack (estimate)	The household's food in the 2027 lean season

The July–September 2026 rains were well below average, and sorghum prices had already doubled between January and August (FAO GIEWS). FEWS NET expects an early lean season and little easing of prices at harvest, so buying at harvest is not a discount this year. The reason to buy then is that grain holds its value while the pound does not: it fell from SDG 8,300 to more than SDG 10,000 per dollar in three weeks of October. A full sack is beyond many households, so buy together with neighbours in your square. Dry the grain, keep it in a sealed bag or drum against weevils, check it monthly, and store it out of sight.

5 · Money: before relatives send money, agree the day's rate and the cash-out fee

* Suggested formulation: check with a health worker before sharing.

CONFIRMED ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
Every transfer	None	What a remittance actually buys

Money sent from relatives abroad is, in practice, this district's insurance. The street rate moves every week, and agents who turn a Bankak balance into cash charge 10–50% where cash is scarce. Compare the rate quoted with that day's parallel rate, ask the fee first, and pay traders by Bankak transfer where they accept it rather than cashing out.

6 · Heat: plan for Ramadan and for April to June, not for this winter

PARTIALLY CONFIRMED FROM ACTION 8

BY WHEN	COST TO YOU	PROTECTS
Set up by 1 Feb 2027; Ramadan about 8 Feb – 9 Mar	None	Older and chronically ill people, outdoor workers

December to February carries almost no heat risk here: Climate Impact Lab forecasts about zero extra heat deaths in those months for Khartoum State. Typical highs are about 37°C in March and 40°C in April, and the dangerous heat comes in April to June. Choose the coolest shaded room in the house, do outdoor work such as rebuilding and rubble removal before 10:00, and agree a daily noon check with one neighbour for anyone old or living alone. During Ramadan, drink water and take salt at iftar and suhoor. Islamic rulings allow the sick to break the fast; ask your imam if someone becomes dizzy or confused.

7 · Chronic illness: find the nearest working dialysis unit now, and keep two weeks of insulin cool*

PARTIALLY CONFIRMED FROM ACTION 7

BY WHEN	COST TO YOU	PROTECTS
30 Nov 2026	None to ask; insulin at your usual cost	People on dialysis or insulin

About 3,800 dialysis patients died in Sudan in the war's first year, according to the head of the national kidney centre. Khartoum State reopened 15 dialysis centres on 23 February 2026 but did not publish where they are, so ask the Bahri Emergency Response Room or

* Suggested formulation: check with a health worker before sharing.

Bahri Teaching Hospital which one serves Kafouri, and how to get there when fuel is short. Keep two weeks of insulin cool through power cuts in a clay-pot cooler (zeer) or wrapped in a wet cloth in the shade.*

8 · Explosives: do not touch, move, or burn metal objects, and do not clear rubble yourself; mark the spot and report it

CONFIRMED ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
Now	None	Children above all

The UN Mine Action Service has found anti-vehicle mines in Bahri. Explosive remnants killed or injured 23 people in Khartoum State in January–March 2026, 21 of them children (AP, citing UN data). Teach children not to pick anything up. Report items to the Sudan National Mine Action Center on 60666; we could not confirm that the number still works in 2026, so ask the Emergency Response Room for the current one.

9 · Power: keep any generator outdoors and away from windows, never in a closed room or courtyard shed*

PARTIALLY CONFIRMED ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
Now	None	Everyone asleep in the house

Generator exhaust contains carbon monoxide, which has no smell and kills people in their sleep. We found no recorded case in Sudan, but generator fumes killed nine members of one family in Tripoli, Libya, and a mother and five children in Dakahlia, Egypt. With outages of 8–10 hours a day in Bahri, more households are running generators this year.*

What to ask of the people in charge

- Ask the **Bahri Emergency Response Room** whether there is an oral rehydration point in Kafouri, and who the volunteer for your square is (action 1).

* Suggested formulation: check with a health worker before sharing.

- Ask the **WASH cluster partner serving Bahri** when chlorine tablets will next be handed out in Kafouri (action 2).
- Ask the **Khartoum State Ministry of Health** whether suspected cholera cases from Kafouri and Al-Ezba are being tested, so that an emergency vaccine request can be made within seven days of an outbreak being confirmed (action 3).
- Ask your **mosque committee** to back a household count that records no names, so that older people living alone can be checked on in the heat (action 8).

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before the peak and before the April heat. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet, but the evidence behind each was checked for whether it came from somewhere like here.

1. Put an oral rehydration point within walking distance of every square in Kafouri, with one volunteer per square who carries ORS and reports watery diarrhoea to the Bahri Emergency Response Room the same day

PARTIALLY CONFIRMED | EMERGENCY MANAGER · PUBLIC HEALTH · FRONTLINE · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Bahri Emergency Response Room	30 Nov 2026	Within SDG 352 million (US$40,000), shared with action 2	Cholera and diarrhoea deaths at home, at night, after the salts run out

ORS may prevent about 93% of diarrhoea deaths in children under five (Munos et al., *International Journal of Epidemiology* 2010, a review of mostly low-quality studies). Khartoum State recorded more than 16,000 cholera cases and 239 deaths by 12 June 2025, and in August 2026 the doctors' union reported uncounted cases in Al-Ezba, next to Kafouri. The Bahri Emergency Response Room already publishes monthly disease counts; same-day reporting would be a new job for it. ORS sachets cost about SDG 530 (US$0.06) each (UNICEF catalogue), so the cost is the volunteers and the sites, not the salts. Same-day counts are also what the Ministry of Health needs to confirm an outbreak and order vaccine (action 3).

* Suggested formulation: check with a health worker before sharing.

Transferability: the ORS evidence is from children in low-income settings and transfers well; the reporting design is untested here, and people who stayed under the RSF may avoid anything that looks like a list (see "Do not do this").

2. Chlorinate at every water point, tanker, and tank serving Kafouri, and hand out household chlorine tablets before the next long power cut

PARTIALLY CONFIRMED | EMERGENCY MANAGER · PUBLIC HEALTH · SKEPTIC · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
WASH cluster partner for Bahri, with the Bahri Emergency Response Room testing free chlorine	**Start now; stock in homes by 15 Dec 2026**	**Within SDG 352 million (US$40,000), shared with action 1**	**Water-borne disease while the Bahri plant is only partly running**

Treating water at the point of use cut child diarrhoea by about 34% (risk ratio 0.66, 95% CI 0.56–0.77) across 25 studies in low- and middle-income countries (Wolf et al., *Lancet* 2022); the effect is smaller in blinded trials, and the outcome is all diarrhoea, not cholera deaths. Across 18 trials, water treatment cut deaths in children under five by about a quarter (Kremer et al., NBER 2023), and a Kenyan chlorine-dispenser programme cost about SDG 17 million (US$1,941) per death averted (Haushofer et al.). The Bahri plant, designed for 300,000 m³ a day, has only partly resumed, and in 2025 drone strikes on power stations stopped the pumps and sent residents back to Nile water. The WASH cluster's target is 0.2–0.5 mg/L of free chlorine after 30 minutes.

* Suggested formulation: check with a health worker before sharing.

Transferability: good; the trials are from low-income settings with unsafe supply. Feasibility is high, because no new authority is needed, only tablets, bleach, and test kits.

3. Be ready to vaccinate: test suspected cholera from Kafouri and Al-Ezba now, file an emergency vaccine request within seven days of a confirmed outbreak, and plan a mop-up round for returnees

PARTIALLY CONFIRMED | EMERGENCY MANAGER · PUBLIC HEALTH · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Khartoum State Ministry of Health, with WHO and UNICEF	Testing now; request within 7 days of confirmation	Vaccine from the global stockpile; delivery about SDG 6,000–8,500 (US$0.68–0.97) a dose	A repeat of the 2025 wave among people who arrived after the last round

Bahri was left out of the June 2025 campaign but was covered in a later 2025 round, which reached 641,851 people in Bahri, Karari, and Umbadda against a target of 1,202,381 (Sudan News Agency, September 2025, an interim count). Many residents have returned since. Emergency doses from the International Coordinating Group stockpile need a confirmed outbreak and a request within seven days of confirmation; preventive doses through the global cholera task force are unlikely to arrive before March 2027. One dose was about 87% effective in the short term in Juba (Azman et al., *Lancet Global Health* 2016, 34 confirmed cases). Published delivery costs, excluding vaccine, are SDG 6,000–8,500 (US$0.68–0.97) a dose (Ethiopia 2015, Malawi 2016); for a 50,000-person planning catchment that is about SDG 300–427 million (US$34,000–48,500).

* Suggested formulation: check with a health worker before sharing.

Transferability: Juba is a comparable conflict city. We found no confirmed cholera outbreak in Khartoum in September–October 2026, which is why the action is readiness, not a campaign.

4. Run a weekly container sweep in every square from November to March, and target malaria nets at the households most at risk; do not count fogging as control

PARTIALLY CONFIRMED | PUBLIC HEALTH · FRONTLINE · SKEPTIC · HISTORIAN · EMERGENCY MANAGER | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Khartoum State Ministry of Health vector unit, with the Bahri Emergency Response Room and mosque committees	Start 15 Nov 2026	Volunteer time; nets about SDG 17,600 (US$2) each before delivery	Dengue and malaria after the rains and again as it warms

Community-led removal of breeding sites cut dengue infection in children by 29.5% (95% CI 3.8–55.3) in a cluster-randomised trial in Nicaragua and Mexico (Andersson et al., *BMJ* 2015). In Sudan, the dengue mosquito breeds mainly in stored-water containers: about half of drinking-water containers were positive in Port Sudan, and clay pots were the main site in El Geneina. Medics counted at least 186 dengue deaths in Samarab, Bahri, in October 2024, and the Ministry of Health ran its 2025 campaign at 70 sites in Bahri. Nets protect against malaria, including the invasive *Anopheles stephensi* found in Khartoum, which also breeds in tanks; they do not stop dengue, which day-biting mosquitoes carry.

* Suggested formulation: check with a health worker before sharing.

Transferability: moderate. The trial was in Latin America; the breeding-site evidence is Sudanese but not from Khartoum.

5. Buy up to 60 tonnes of sorghum for the Bahri community kitchens in December–January, count local prices weekly, and reset any cash transfer every week

PARTIALLY CONFIRMED | FOOD SUPPLY · HISTORIAN · ECONOMIST · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Bahri Emergency Response Room kitchens, with WFP where it operates	Buy Dec 2026 – Jan 2027	SDG 176–352 million (US$20,000–40,000), priced on the day	Kitchens closing when prices rise in the 2027 lean season

The 2026 rains were below average across most cropping areas, with Gedaref about 30% and Sennar about 20% below the long-term average (FAO GIEWS, 24 September 2026); Reuters, citing USGS data, called July–September the driest in more than 30 years. Sorghum prices had already more than doubled between January and August. In the 2015 El Niño the cereal harvest came in about 30% short and, by May 2016, sorghum cost about 65% more than its five-year average (FEWS NET). This year there is no carryover stock: the 2025 harvest was 22% below 2024. FEWS NET expects an early lean season with "little moderation in prices" at harvest, so the case for buying in December is holding grain against the falling pound, not a seasonal discount. Monthly price series (IFPRI, WFP) run one to two months late and none is Khartoum-only, so the kitchens' own weekly market count is the working trigger.

* Suggested formulation: check with a health worker before sharing.

Transferability: the precedent is Sudanese. Feasibility depends on how many Bahri kitchens are still cooking: Islamic Relief reported 354 of 844 kitchens closed in 2026, and we found no 2026 WFP kitchen programme in Bahri. Confirm the count before committing 60 tonnes.

6. Keep the Bahri Emergency Response Room running: apply now for Sudan Humanitarian Fund mutual-aid grants through an international NGO, and ask for a satellite link and two to four weeks of grain per kitchen cluster

PARTIALLY CONFIRMED | PHILANTHROPIST · FOOD SUPPLY · HISTORIAN · ECONOMIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
OCHA Sudan Humanitarian Fund, through its INGO intermediaries	**Apply in Nov 2026, for funds before the lean season**	**Grants up to SDG 176 million (US$20,000) each with light approval**	**The district's working health and food network collapsing mid-season**

The Emergency Response Rooms are what actually works here, and they are fragile: about 900 of 1,400 kitchens nationally stopped after the January 2025 US aid freeze, and 59 Bahri kitchens stopped during a communications outage in early 2024. The Sudan Humanitarian Fund holds pledges of about SDG 6.2 trillion (US$700 million) for 2026 (UAE US$500 million and US US$200 million, February 2026) and funds mutual-aid groups through six INGO intermediaries, in grants of up to SDG 44 million (US$5,000) with no clearance and up to SDG 176 million (US$20,000) with light approval; direct funding requires registration with the Humanitarian Aid Commission, which many groups avoid. UK FCDO also funds local responders directly. Neither offers twelve months of core costs as a product, so that is an ask, not an existing grant. Whether Bahri is among the Fund's 84 "hyper-prioritised locations" is not published.

* Suggested formulation: check with a health worker before sharing.

Transferability: the mechanism is Sudanese and already in use. Its trigger is discretionary, so the money moves on an application, not a forecast.

7. Map which dialysis unit is serving Kafouri, keep a list of who needs it and insulin, and stock Bahri Teaching Hospital for the patients the rehydration points refer

PARTIALLY CONFIRMED | PUBLIC HEALTH · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Khartoum State Ministry of Health, with Bahri Teaching Hospital	31 Dec 2026	Mapping none; hospital supplies SDG 264 million (US$30,000)	Deaths when dialysis, insulin, or severe-case care stops

About 3,800 dialysis patients died in Sudan from April 2023 to April 2024, according to the head of the National Centre for Kidney Diseases and Surgery; his figures do not add up exactly, so treat it as a reported count. Khartoum State reopened 15 dialysis centres on 23 February 2026 without naming their sites. Bahri Teaching Hospital reopened on about 18 January 2026 with about 70% of it damaged; whether it offers dialysis is not confirmed, and we found no international organisation supporting it. The SDG 264 million (US$30,000) buys intravenous fluids, a cholera treatment corner, and dengue and malaria rapid tests, delivered as supplies through the Ministry and WHO pipeline. No study gives an effect size; the case rests on near-certain death when this care stops.

Transferability: not applicable; this is a service gap, not an intervention with trial evidence. Any list of patients should record need, not where people were during the war.

8. Before the April heat, count who lives alone in each square without recording names, pair each with a neighbour, and brief imams in February for Ramadan and the hot months

PARTIALLY CONFIRMED | FRONTLINE · SKEPTIC · PUBLIC HEALTH | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Bahri Emergency Response Room, cleared with mosque committees	Set up by 1 Feb 2027; run through Ramadan and April–June	Volunteer time	Heat deaths among older people alone in houses without power

* Suggested formulation: check with a health worker before sharing.

Climate Impact Lab forecasts 970 extra heat deaths in Khartoum State in October, about zero from December to February, and 130 in March; typical highs are about 37°C in March and 40°C in April, and the worst heat comes in April to June. Ramadan falls on about 8 February to 9 March 2027, with fasting as temperatures rise. Ahmedabad's heat action plan was associated with about 1,190 fewer deaths a year (95% CI 162–2,218; Hess et al. 2018), but that was a before-and-after study in India that relied on forecasts, hospital readiness, and text warnings Bahri does not have; only the outreach part transfers, and neighbour checks on their own have no measured effect. Sermons reach people who are not on WhatsApp when the network is down. Fans and cooling centres are not the answer here; see "Do not do this".

Transferability: low to moderate. The design is cheap and fits how Kafouri communicates, but the evidence comes from a different city and health system.

9. Sudanese Red Crescent: prepare an IFRC emergency fund request for an April–June heatwave now, so it can be filed the day the Meteorological Authority warns

PARTIALLY CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Sudanese Red Crescent Society, Khartoum branch	**Request drafted by 31 Mar 2027**	**Unfunded until sized; Syria's 2025 heat request was about CHF 170,000**	**Heat response money arriving after the heatwave**

No early action protocol covers Sudan: the Red Crescent's riverine-flood protocol is still listed as under development, and there is none for heat. Sudan's African Risk Capacity drought cover ran only in 2021–22. The IFRC Disaster Response Emergency Fund can still pay out for an imminent crisis. We propose, as our own threshold and not an IFRC rule, a Sudan Meteorological Authority forecast of at least 44°C for three or more days at the Khartoum station: plausible from April to June (the Authority warned of more than 47°C in July 2025) and unlikely in March. Syria's 2025 heat request, the first of its kind, was about CHF 170,000; Bangladesh's heat protocol activation was about CHF 500,000.

Transferability: the instrument is global; no heat request has been made for Sudan, so the size and the trigger are untested here.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	State Ministry of Health	Bahri ERR	WASH cluster	WHO / UNICEF	OCHA SHF	Sudanese Red Crescent	Bahri Teaching Hospital	Mosque committees	WFP	Met Authority
1. Rehydration points and same-day reports	C	A/R	I	C	I	R	C	R		
2. Chlorination and household tablets	C	R	A/R	C	C	R		I		
3. Cholera testing and vaccine readiness	A/R	C	I	R	I	R	C	I		
4. Weekly container sweep and malaria nets	A	R	C	C		R		R		
5. Sorghum for kitchens and weekly prices		A/R			C			C	C	
6. Keep the Emergency Response Room funded	I	R			A				C	
7. Dialysis map, patient list, hospital stock	A	R		C			R			
8. No-names count and neighbour checks	C	A/R				C		R		I
9. Heatwave emergency fund request	C	I			I	A/R				C

What the chart shows: the Bahri Emergency Response Room does the work on eight of the nine rows and is accountable for four, which makes its survival (row 6) the single point of failure. The Ministry of Health is accountable for the three medical rows. Rows 3, 8, and 9 need decisions and preparation more than money.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

SDG 352 million (US$40,000)

TARGET

SDG 968 million (US$110,000)

CEILING

SDG 1.87 billion (US$212,000)

LARGEST UNCERTAINTY

Whether a cholera outbreak is confirmed in Bahri, which decides whether vaccine money is needed

In US dollars: floor = placement 1 = US$40,000. Target = placements 1 + 2 + 3 = 40,000 + 30,000 + 40,000 = US$110,000. Ceiling = (110,000 + 50,000 vaccine-delivery contingency + 10,000 heat) × 1.25 for currency and access risk = US$212,500, rounded to US$212,000. The vaccine contingency is 50,000 people × US$0.97 a dose ≈ US$48,500, rounded up; 50,000 is a planning assumption, because Kafouri's population is

* Suggested formulation: check with a health worker before sharing.

not published. The target fell from the panel's US$195,000 (–44%): the vaccine round moved to the ceiling because doses are released only for a confirmed outbreak, and the sorghum was re-priced at SDG 176–352 million (US$20,000–40,000) for 60 tonnes.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Bahri Emergency Response Room, through an SHF intermediary	Actions 1 and 2: chlorine at water points and tankers, household tablets, two rehydration points, square volunteers	SDG 352 million (US$40,000), in two tranches of SDG 176 million (US$20,000)	Direct grant	Water treatment cut under-5 deaths by about a quarter (Kremer et al. 2023, 18 trials); about SDG 17 million (US$1,941) per death averted (Kenya dispensers, Haushofer et al.); ORS about SDG 530 (US$0.06) a sachet (UNICEF)	Partially confirmed
Bahri Teaching Hospital, through the Ministry and WHO supply pipeline	Action 7: intravenous fluids, a cholera treatment corner, dengue and malaria rapid tests	SDG 264 million (US$30,000)	Pre-positioned stock	Severe cases need somewhere to go once rehydration points refer them; absorptive capacity unverified	Partially confirmed
Bahri Emergency Response Room kitchens	Action 5: up to 60 tonnes of sorghum and 2–4 weeks of grain per kitchen cluster	SDG 352 million (US$40,000)	Pre-positioned stock	Sorghum about SDG 3,000–4,000 a kg (estimate; no Khartoum price published); weakest link to deaths averted	Partially confirmed
Khartoum State Ministry of Health, with WHO and UNICEF	Action 3: delivery of a mop-up or outbreak vaccine round (ceiling only)	SDG 440 million (US$50,000)	Top-up	SDG 6,000–8,500 (US$0.68–0.97) a dose delivered, vaccine excluded (Ethiopia 2015, Malawi 2016); released only for a confirmed outbreak	Unverified
Bahri Emergency Response Room	Action 8: shade and drinking water at kitchens and clinics, April–June (ceiling only)	SDG 88 million (US$10,000)	Direct grant	No published cost-effectiveness; heat burden here is small until April	Unverified

* Suggested formulation: check with a health worker before sharing.

The funder's one thing: the cheapest lives here are bought with chlorine and oral rehydration salts, not with heat programmes. The constraint is not the money but whether the Emergency Response Room still exists next quarter. We found no Khartoum or Bahri line for water, health, or food in the Sudan Humanitarian Fund's 2026 allocations, OCHA's June 2026 grant, or the 2026 response plan, which was about 20% funded in April.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here, each checked against a documented case.

Do not clear rubble, move metal objects, or encourage families back into a block before it has been checked for explosives

CONTRAINDICATED

The UN Mine Action Service reported anti-vehicle mines in Omdurman and Bahri in August 2025. In January–March 2026, 23 people were killed or injured by explosive remnants in Khartoum State, 21 of them children (AP, citing UN data); the national figure for the quarter was 77. IOM warns that nearly four million returns risk becoming unsustainable without services.

Instead: explosive-risk education first, especially for children, and report suspect items to the Sudan National Mine Action Center (60666; confirm the number locally).

Do not keep lists that record who stayed in Kafouri under the RSF

CONTRAINDICATED

After the March 2025 recapture, UN High Commissioner for Human Rights Volker Türk condemned summary killings of suspected RSF collaborators and said lists of suspected collaborators were circulating; extrajudicial killings were alleged in Bahri. Volunteers have been arrested, and a Khartoum State directive of 12 May 2025 told aid groups to register or be treated as operating illegally. A household that thinks it is being listed will not open the door, and the sick person inside goes uncounted.

* Suggested formulation: check with a health worker before sharing.

Instead: count households, not people, with no names and no history, and clear the plan with the mosque committee first.

Do not build the plan on piped water, grid power, fans, or cooling centres

CONTRAINDICATED

Bahri residents report 8–10 hours without power a day, and the Ministry of Energy said in August 2026 that some areas lose power for more than 18 hours; one grocer in southern Khartoum reported 48 hours. When drone strikes cut the pumps in May 2025, residents went back to Nile water and donkey-cart barrels, and Khartoum recorded more than 1,000 cholera cases a day. A fan or a cooling centre that depends on the grid fails exactly when it is needed.

Instead: household chlorination stocked before the cut (action 2), and shade, ventilation, water, and a neighbour's check at home (action 8).

Do not count indoor fogging as dengue control

CONTRAINDICATED

Fogging kills adult mosquitoes for a short time and goes where cases already are: the Federal Ministry of Health reported 99% indoor fogging coverage in Khartoum State in 2025, and the state still carried 80% of Sudan's 3,676 dengue cases. Those two numbers do not prove fogging fails, but they show it did not stop the outbreak. The larvae in stored water are untouched by it.

Instead: the weekly container sweep, with lids and scrubbing (action 4).

Do not pay aid in cash where agents charge 10–50% to cash out

CONTRAINDICATED

After the December 2024 banknote swap, cash and food distributions for more than four million people were delayed by over a month. Where banknotes are scarce, agents take 10–50% to turn a Bankak balance into cash (CGAP), and a Bankak outage in May 2026 brought Khartoum markets to a near halt.

* Suggested formulation: check with a health worker before sharing.

Instead: pay by transfer to traders who accept Bankak, give grain where they do not, check prices weekly, and reset transfer values weekly as the pound falls.

Do not run a generator indoors, in a closed courtyard shed, or by a window*

CONTRAINDICATED

We found no recorded case in Sudan, but in comparable places generator fumes killed nine members of one family in Tripoli, Libya, and a mother and five children in Dakahlia, Egypt. With daily outages, more households in Bahri are running generators this year.

Instead: run it outdoors, away from doors and windows, and never while people sleep in a closed room nearby.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1984–85	Drought and famine after two failed rainy seasons; migrants pour into Khartoum	About 240,000 dead nationally (estimate, de Waal), about 105,000 in Darfur	Price protests and a general strike in Khartoum ended in Nimeiry's overthrow on 6 April 1985
4–5 Aug 1988	Khartoum flood: 210 mm of rain in 24 hours, in a La Niña year after the 1986–87 El Niño	About 80 dead (58 reported by 16 August); 127,000 dwellings destroyed; about 750,000 people affected	A large international airlift of relief
2015–16	El Niño: rains 25–80% below normal in the east	Cereal harvest about 30% below average; sorghum about 65% above its five-year average by May 2016; more than 4.4 million people in crisis or worse	Carryover stocks softened the price rise by mid-2016
Aug–Sep 2020	Blue Nile reaches a record 17.57 m at Khartoum (NASA gives 17.4 m)	About 100 dead nationally	State of emergency; the drainage network promised in 2014 was still unfinished

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Oct 2024	Dengue in Samarab, Bahri, then under RSF control	At least 186 deaths (medics' count), about 500 cases	Ministry campaign at 70 Bahri sites in 2025
2025	Cholera in Khartoum State after power cuts stopped the water pumps	More than 16,000 cases and 239 deaths in the state by 12 June; up to 1,500 new cases a day in late May	Vaccination: 2.24 million people in five localities in June, not Bahri; Bahri covered in a later round (641,851 of 1,202,381 targeted across three localities)
Jul–Sep 2026	El Niño rainy season fails: Gedaref about 30% and Sennar about 20% below average	Sorghum prices already more than doubled January–August	Lean season expected early in 2027

The recurrence nobody has noticed: in Khartoum the floods have tended to come after the El Niños, not during them. The 1986–87 El Niño was followed by the 1988 flood and the 1997–98 El Niño by the 1998 flood, each in a La Niña year; 2020 fits only loosely, and the 1988 flood was local rain while 2020 was the Nile. It is a pattern of two clean cases and one borderline, not a law, but on it this El Niño brings a hungry 2027 and the drains promised in 2014 should be cleared before the 2027 and 2028 rains.

WHERE THE EXPERTS DISAGREE

UNRESOLVED

- **"El Niño is a rounding error here"** versus **"El Niño's drought will drive food prices through 2027"**. The Skeptic notes that only 1,100 heat deaths remain forecast for the whole state, mostly in an October that is already ending, and that an El Niño label could pull money from chlorine and demining. The food supply analyst points to the worst rains in decades and a 2015–16 precedent of sharply higher prices. Both hold: the heat signal is small, and the food signal is real but late. The 2027 lean-season price series will settle it.
- **Cooling points with solar fans at Emergency Response Room sites** (Emergency manager) versus **no cooling centres or fans in a city without reliable power** (Skeptic). We ranked home-based shade and neighbour checks, and left shade and water at kitchens in the ceiling. Evidence that would settle it: any evaluation of solar-powered cooling sites in a Sahelian city.
- **Cap the commission on cash-out agents** (Economist) versus **a cap would push the trade off the books** (verification). No Central Bank of Sudan rule caps it, and the commission reflects a real shortage of banknotes. The proposal could not be verified and was dropped from the ranked actions.

* Suggested formulation: check with a health worker before sharing.

WHAT NOBODY COVERED

BLIND SPOT

No lens covered the treatment of acute malnutrition in children, which is how a food-price shock turns into deaths: if Bahri has no working outpatient therapeutic feeding site, the price rise in action 5 lands on children with nowhere to go. Nor did any lens cover protection from gender-based violence or mental health care for returnees. And Kafouri's population is not published, so every per-person cost here rests on a 50,000-person planning assumption; a household count (action 8) would replace it.

HOW TO READ THIS

METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

The Climate Impact Lab figure here is published for Khartoum State as a whole, at province level; it is not a neighbourhood figure. The El Niño projection covers *heat only*; cholera, dengue, malaria, hunger, and flood are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was taken as given: Kafouri, a residential district of Khartoum North in Bahri locality, Khartoum State, divided into numbered squares 1–12; no population is published. Climate Impact Lab figures are available only for Khartoum State (province level); no impact region was resolved. Sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab El Niño Heat Mortality Tracker data, file `v261026_adm1_output_BASE_rounded_cust.csv` (version of 26 October 2026, pre-peer-review): Khartoum State, 6,200 additional heat deaths June–September 2026, 1,100 forecast October 2026 – March 2027 (October 970, November 45, December −14, January 24, February −41, March 130), 7,300 in total; Sudan, 53,200; the world, 857,400. This update supersedes the September report *451,000 at Risk*. Method: ECMWF SEAS5 seasonal forecasts applied to temperature–mortality relationships against a 1996–2025 baseline. The figures count heat only; we inferred this from the data, because cold countries show exactly zero in every winter month. They are never added to the Impact Map's long-run projections.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources in English and

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Arabic, to a fixed brief.

- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in Sudan, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. Food actions were checked for whether each price series is actually published, the timing of the price peak, and the kitchens' capacity.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not consulted. Its "Download Inventory" link leads to a registration form, and the browse page loads its data in the browser. No action here should be read as having no precedent; none was checked.
- **Money.** Costs are shown in Sudanese pounds with US dollars in parentheses, at SDG 8,800 = US$1, the parallel-market rate reported on 7 October 2026 (Rio Times). We used the parallel rate because households, traders, and aid transfers all price at it; banks quoted about SDG 3,750–4,200 on 17 September 2026, and the parallel rate passed SDG 10,000 by 21 October, so SDG figures here will understate costs as the pound falls. Most source costs are in US dollars, which humanitarian budgets in Sudan use.
- **Reader.** This briefing was built at El Niño Ready's initiative, not from a request. No role was given, so the For you section is written for a resident. The "For you" steps were taken from the verified actions and from what the panel found about residents here; any step not drawn from a ranked action was verified on the same three tests. The rest of the report is written for the State Ministry of Health, the Bahri Emergency Response Room, and their partners.
- **Health wording.** Passages where a wording error could hurt someone (water treatment, oral rehydration salts, chronic illness and dialysis, and generators and carbon monoxide) are marked "* Suggested formulation: check with a health worker before sharing." They were checked

* Suggested formulation: check with a health worker before sharing.

against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.

- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Sources on Kafouri itself are almost absent; most figures are for Bahri or Khartoum State. Several Sudan Tribune and allAfrica pages returned errors, so some figures (the later 2025 vaccination round in Bahri, the 21 October exchange rate) rest on search excerpts. Some Sudanese news pages carry render dates rather than publication dates, so undated items were treated as unverified. No Bahri retail prices are published, so household costs are estimates. The locator map shows Kafouri at country scale; the district itself is not drawn.

References · status after verification

Climate Impact Lab, El Niño Heat Mortality Tracker data v261026 (downloaded 30 Oct 2026); tracker · confirmed

Cholera, Khartoum 2025 and vaccination: WHO EMRO, June 2025; WHO, cholera in all but one state; SUNA via allAfrica, Bahri round · corrected (excerpts); UNICEF, Khartoum State; Al-Ezba, Aug 2026: EMPHNET · confirmed; ICG stockpile rules: WHO · confirmed

Vaccine effect: Azman et al. 2016, Juba · confirmed; delivery costs: Ethiopia, Malawi · confirmed

Water treatment: Wolf et al., Lancet 2022; Kremer et al., NBER w30835; Haushofer et al., Kenya; bleach dosing: US CDC · corrected; Bahri water plant: Sudan Tribune · corrected

ORS: Munos et al. 2010; UNICEF supply catalogue; home recipe: Rehydration Project · corrected

Dengue and malaria: Andersson et al., BMJ 2015; Port Sudan containers; An. stephensi in Khartoum; Samarab deaths, EMPHNET; fogging coverage, allAfrica; net price: Against Malaria Foundation · corrected

Dialysis: Sudan Tribune, kidney centre; 15 centres reopened; Bahri Teaching Hospital: Africanews/AFP · corrected

Heat: Hess et al. 2018, Ahmedabad; Khartoum climatology; IFRC heat request: Syria 2025; Bangladesh; Sudan anticipatory action: Anticipation Hub · corrected

Funding: SHF update, April 2026; UAE pledge; Refugees International on localisation; UK FCDO; OCHA grant, June 2026 · corrected

Food: FAO GIEWS Sudan, 24 Sep 2026; FEWS NET, Sep 2026; FEWS NET, June 2016; FEWS NET Price Watch, July 2016; Reuters via CNBC Africa, 23 Oct 2026; IFPRI, July 2026; Islamic Relief, kitchens; WFP Sudan; Emergency Response Rooms, 2024 outage · corrected

Money: Rio Times, 7 Oct 2026 · confirmed; Sudan Tribune, 21 Oct 2026 · excerpts; bank rates, 17 Sep 2026 · confirmed; cash-out: CGAP; OCHA, January 2025 · corrected

Explosives and protection: UNMAS Sudan; AP via Africanews, 29 Apr 2026; IOM on returns; The New Humanitarian, volunteers targeted; ERR oversight order · confirmed

Power and water: Africanews, June 2025; MSF, 6 Oct 2026; Andariya, services and wages · corrected

History: de Waal, LRB; MERIP 1985; 1988 floods; Reuters, 16 Aug 1988; NASA, 2020; Dabanga, drainage; NOAA ONI; Kafouri: Wikipedia · corrected

* Suggested formulation: check with a health worker before sharing.

Exposure data · Climate Impact Lab, El Niño Heat Mortality Tracker · Impact Map · Adaptation Inventory

Generated 30 October 2026 for Kafouri, Khartoum North. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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